

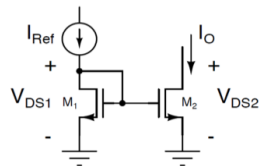
Lecture-17

Revanth Reddy
Pannala
EBIT, Unibo

continuation of current mirror.

MOSFET Mirror: Inaccuracies

Channel-length modulation causes I_D to be a function of V_{DS} .
The scale factor becomes a function of V_{DS} .



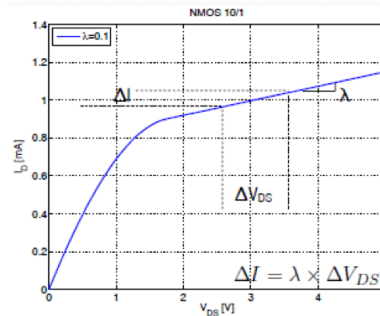
$$I_{Ref} = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{OV}^2 (1 + \lambda V_{DS1}), \quad I_O = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{OV}^2 (1 + \lambda V_{DS2})$$

$$\Rightarrow \varepsilon = \frac{\Delta I}{I_{Ref}} = \frac{I_O - I_{Ref}}{I_{Ref}} = \frac{\lambda(V_{DS2} - V_{DS1})}{1 + \lambda V_{DS1}} \approx \lambda \Delta V_{DS}$$

For $\lambda = 0.1V^{-1}$ and $\Delta V_{DS} = 1V$ the error will be 10%!!

To reduce the error, we can

- Use a device with smaller λ (larger L , $\lambda \propto L^{-1}$)
 - It's common to see non-minimum L in bias circuits.
- Minimize ΔV_{DS}

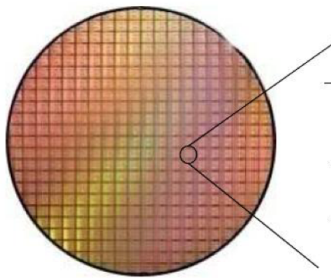


EK

9

- Device parameters not only vary from lot-to-lot or wafer-to-wafer, but there are also differences between closely spaced, nominally identical devices on the same chip
 - These differences are called mismatch

Inaccuracy due to Random Mismatch



$$V_{t1} - V_{t2} = \Delta V_t$$

$$\left(\mu C_{ox} \frac{W}{L} \right)_1 - \left(\mu C_{ox} \frac{W}{L} \right)_2 = \Delta \beta$$



$$C_1 - C_2 = \Delta C$$



$$R_1 - R_2 = \Delta R$$

Random variations in

$$V_t$$

$$\beta = \mu C_{ox} \frac{W}{L}$$

Experiments over the past decades have shown that device-to-device mismatch (ΔV_t , ΔC , ...) for properly laid out devices (no systematic errors) is "random" and well-described by a Gaussian distribution

- With zero mean and a standard deviation that depends on the process and the size of the device

Empirically, the standard deviation of the mismatch between two closely-spaced devices can be modeled using the following expression:

$$\sigma_{\Delta P} = \frac{A_P}{\sqrt{WL}}$$

where WL represents the area of the device, A is a constant and P is the device parameter under consideration.

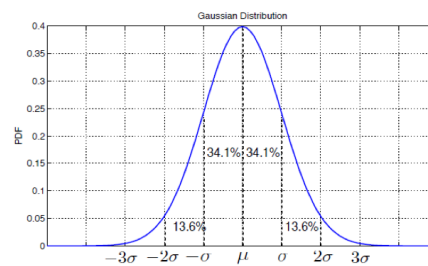


Sample technology

Parameter	Value
A_{V_t}	20mV. μ m
$A_{\frac{\Delta\beta}{\beta}}$	2%. μ m

For a MOSFET with $W=20\mu\text{m}$ and $L=1\mu\text{m}$:

$$\sigma_{\Delta V_t} = \frac{20}{\sqrt{20}} = 4.5\text{mV}, \quad \sigma_{\frac{\Delta\beta}{\beta}} = \frac{2\%}{\sqrt{20}} = 0.45\%$$



11

Propagation of uncertainty

Any non-linear differentiable function, $f(a,b)$, of two variables, a and b , can be expanded as

$$f \approx f^0 + \frac{\partial f}{\partial a}a + \frac{\partial f}{\partial b}b$$

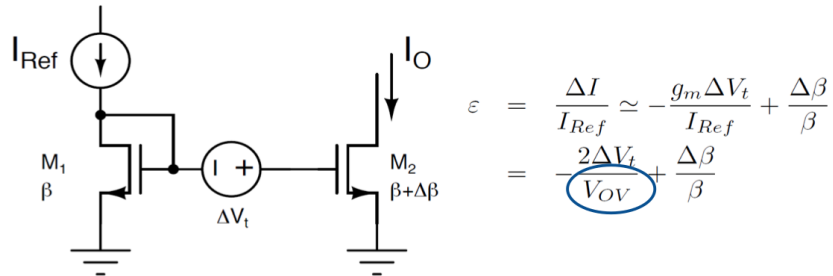
hence:

$$\sigma_f^2 \approx \left| \frac{\partial f}{\partial a} \right|^2 \sigma_a^2 + \left| \frac{\partial f}{\partial b} \right|^2 \sigma_b^2 + 2 \frac{\partial f}{\partial a} \frac{\partial f}{\partial b} \sigma_{ab}.$$

Function	Variance	Standard Deviation
$f = aA$	$\sigma_f^2 = a^2 \sigma_A^2$	$\sigma_f = a \sigma_A$
$f = aA + bB$	$\sigma_f^2 = a^2 \sigma_A^2 + b^2 \sigma_B^2 + 2ab \sigma_{AB}$	$\sigma_f = \sqrt{a^2 \sigma_A^2 + b^2 \sigma_B^2 + 2ab \sigma_{AB}}$
$f = aA - bB$	$\sigma_f^2 = a^2 \sigma_A^2 + b^2 \sigma_B^2 - 2ab \sigma_{AB}$	$\sigma_f = \sqrt{a^2 \sigma_A^2 + b^2 \sigma_B^2 - 2ab \sigma_{AB}}$

12

Inaccuracy due to Random Mismatch

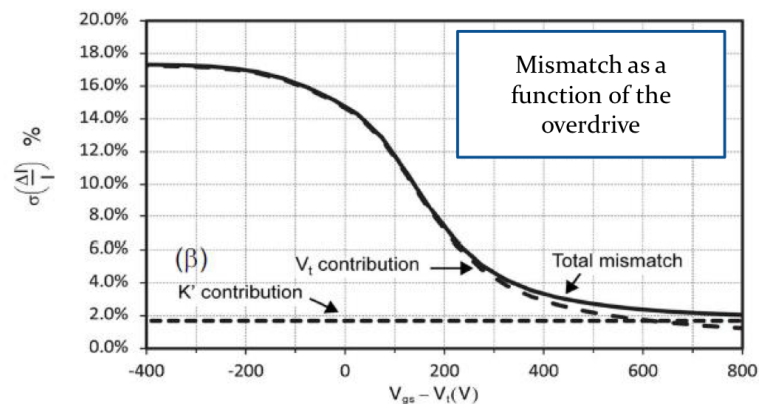


- For $V_{OV}=200\text{mV}$, $W=20\mu\text{m}$ and $L=1\mu\text{m}$:

$$\sigma_\varepsilon = \sqrt{\left(2 \frac{4.5}{200}\right)^2 + (0.45)^2} = 4.5\%$$

13

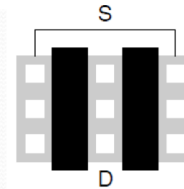
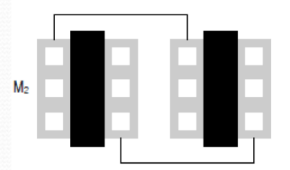
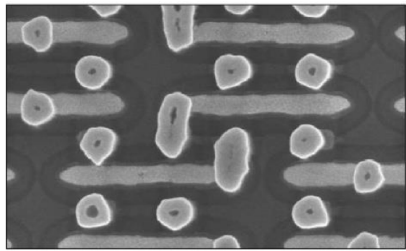
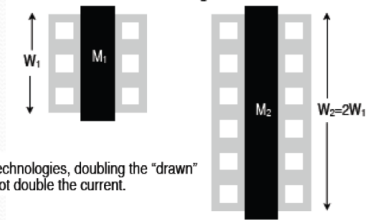
Random Mismatch



14



Scaling Current using W

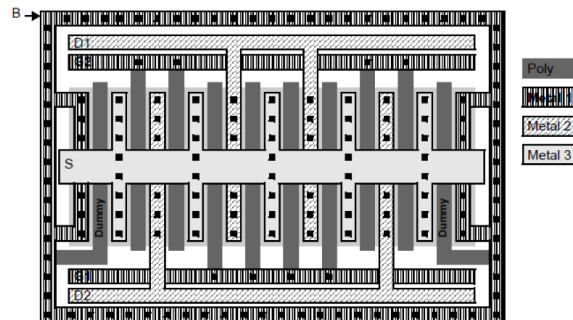
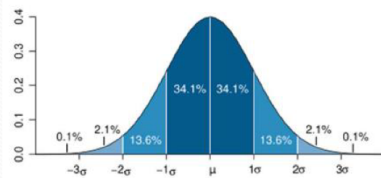
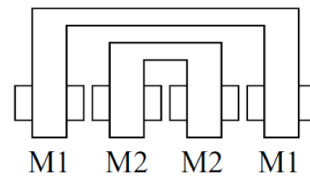
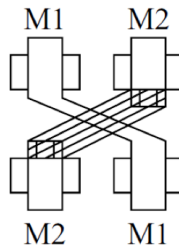
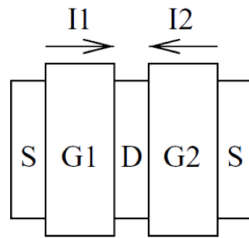


We can make this layout more compact by sharing source and drain regions
- Sharing reduces junction capacitances too.

EK

15

Device Layout: Fingered and Common Centroid

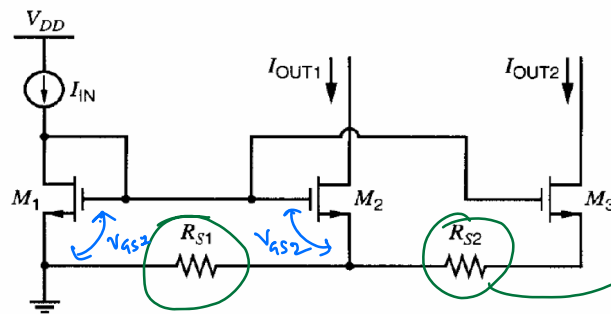


16

తేవంబేరెడి పాపం
EBIT, Unibo



Voltage routing



Goal

$$I_{out1} = I_{IN}$$

$$I_{out2} = I_{IN}$$

It's not exact
because of parasitic
resistances.

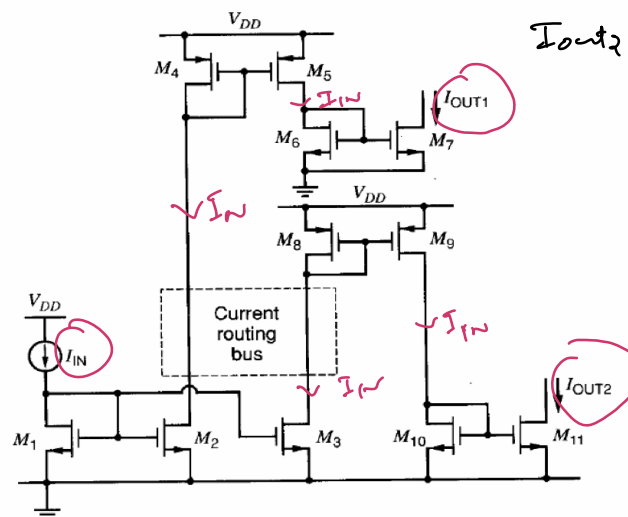
parasitic
Resistance

Not effective when devices to be
mirrored are far apart. Because
 $V_{GS1} \neq V_{GS2}$

21

→ The problems due to voltage routing are solved
using current routing.

Current routing



Goal

$$I_{out1} = I_{IN}$$

$$I_{out2} = I_{IN}$$

22

* The Trick in the case of current Routing is to have closely placed elements of current Mirror and move along the longer path by Transferring information in terms of current.

In the above figure Repeatedly using of current Mirror concept helps in reducing the errors in the current at the output wrt to the input current.

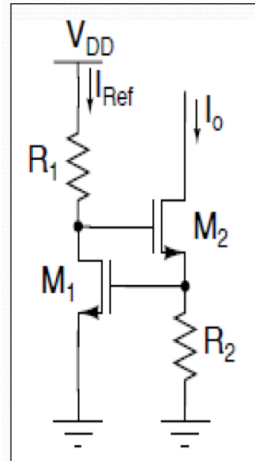
→ Let's investigate ^{other} issues relating to current mirrors & current generators -

Voltage and Current References: dealing with non-idealities

- Conditions to be satisfied for an IC in production:
 - Work even when V_{cc} changes (Supply variation):
 - eg: V_{cc} : 2.7V $\rightarrow$ 3.0V
 - Work even when temperature changes (Temperature variation):
 - eg: T : -25C $\rightarrow$ 0 $\rightarrow$ 25C $\rightarrow$ 75C
 - Work even when physical properties change (Process variation):
 - BJTs: β : $\pm 30\%$
 - MOS: μ : $\pm 10\%$, V_{th} : $\pm 100\text{mV}$
 - Resistors: R : $\pm 20\%$
 - Capacitors: C : $\pm 5\%$
 - Inductors: L : $\pm 1\%$
- All combinations of supply voltage (V_{cc}), temperature (T) and process (P) variations have to be considered in design. This is often referred to as **PVT (process, voltage and temperature)**



Sources Using V_t Standards



Can reduce sensitivity by making I_o depend on voltages other than V_{DD} , e.g., V_t .

$$I_o = \frac{V_{GS1}}{R_2} = \frac{V_t + V_{OV1}}{R_2} = \frac{V_t + \sqrt{\frac{2I_{Ref}}{\mu C_{ox}(\frac{W}{L})_1}}}{R_2}$$

If I_{Ref} is small and $(W/L)_1$ is sufficiently large $V_{OV1} \ll V_t$:

$$I_o \simeq \frac{V_t}{R_2}$$

Therefore, this circuit is known as a *threshold-referenced* bias circuit.

$$\begin{aligned} S_{V_{DD}}^{I_o} &= S_{I_{Ref}}^{I_o} \times S_{V_{DD}}^{I_{Ref}} \\ &\simeq \frac{I_{Ref}}{I_o} \times \frac{\partial I_o}{\partial I_{Ref}} \times \underbrace{S_{V_{DD}}^{I_{Ref}}}_{\simeq 1} \\ &\simeq \frac{1}{2} \frac{V_{OV1}}{V_t + V_{OV1}} \end{aligned}$$

This sensitivity not exactly one-

For example, if $V_t = 1$ V, $V_{ov1} = 0.1$ V, and $S_{V_{DD}}^{I_{IN}} \simeq 1$

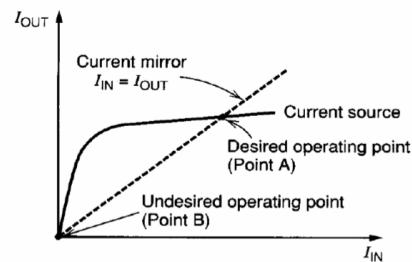
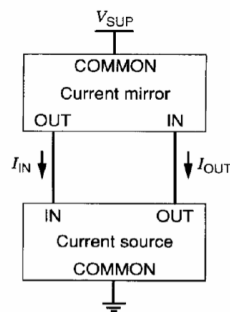
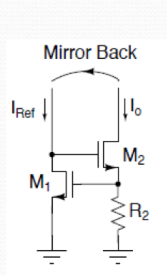
$$S_{V_{DD}}^{I_{OUT}} \simeq \frac{0.1}{2(1.1)} \simeq 0.045$$

GM-EK

24

There are other sophisticated techniques to achieve low sensitivity: which we will discuss now.

Combining Current Sources and Current Mirrors

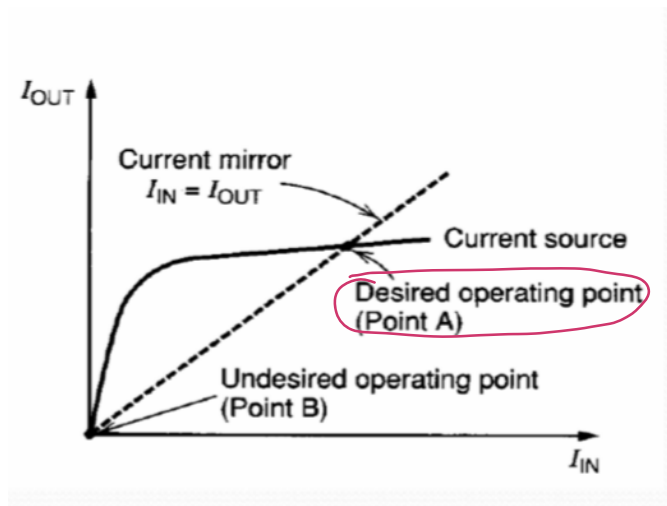
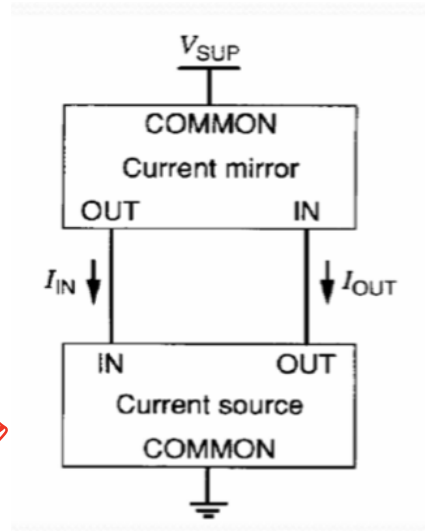
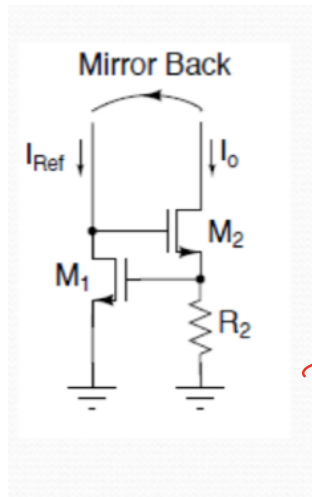


- In V_t -Referenced current source, I_{Ref} is still a function of V_{DD} thus making the circuit sensitive to V_{DD} .
- If everything works out fine I_o will be a replica of I_{Ref} .
- We can mirror back I_o to input instead of using supply-dependent I_{Ref} .

HEK

25

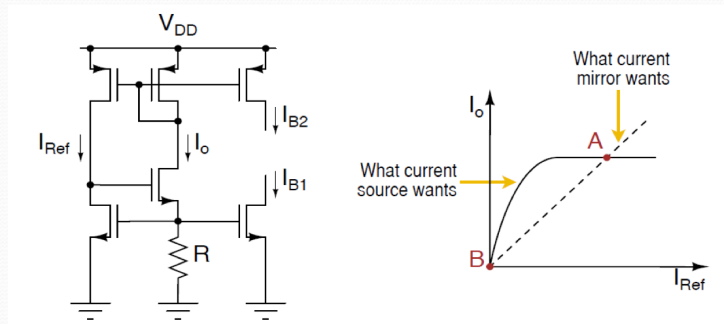
The idea is to start from the V_T Reference Configuration Topology and then Mirror back the o/p current of this configuration into i/p one.



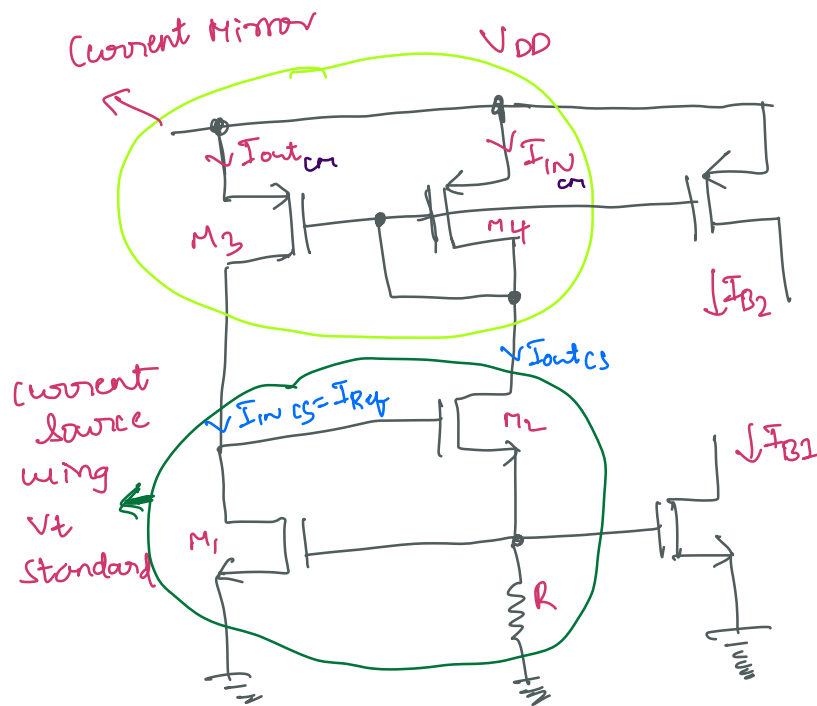
✱✱

let's see how does this system work in practice

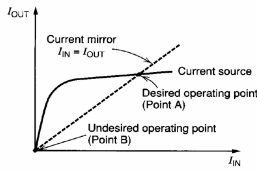
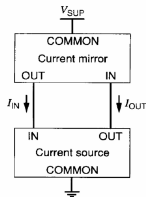
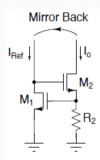
2 stable points



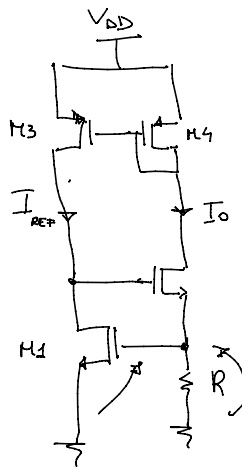
26



Combining Current Sources and Current Mirrors



- In V_T -Referenced current source, I_{ref} is still a function of V_{DD} thus making the circuit sensitive to V_{DD} .
- If everything works out fine I_o will be a replica of I_{ref} .
- We can mirror back I_o to input instead of using supply-dependent I_{ref} .

Size of $M_3 = \text{Size of } M_4$

$$S_3 = S_4$$

$$V_{GS1} = R \cdot I_O = V_T + V_{ov1} = V_T + \sqrt{\frac{2 I_{REF}}{\beta_1}}$$

$$R \cdot I_O = V_T + \sqrt{\frac{2 I_{REF}}{\beta_1}} \quad (1)$$

$$I_{REF} = I_O \quad (2)$$

Because V_{GS} of transistors M_3 & M_4 are equal.

From (1) & (2)

$$I_O - \frac{1}{R} \sqrt{\frac{2 I_O}{\beta_1}} - \frac{V_T}{R} = 0 \quad (3)$$

→ This is a second order equation.

2nd for $\sqrt{I_O}$

$$\sqrt{I_O} = \frac{\frac{1}{R} \cdot \sqrt{\frac{2}{\beta_1}} + \sqrt{\frac{1}{R^2} \cdot \frac{2}{\beta_1} + \frac{4 V_T}{R}}}{2} = \frac{1}{R} \left(\frac{1}{\sqrt{\beta_1} \cdot 2} + \sqrt{\frac{1}{\beta_1 \cdot 2} + V_T \cdot R} \right)$$

$$I_O = \frac{1}{R^2} \cdot \left[\frac{1}{\beta_1 \cdot 2} + \frac{1}{\beta_1 \cdot 2} + V_T \cdot R + 2 \sqrt{\frac{1}{\beta_1^2 \cdot 4} + \frac{V_T \cdot R}{\beta_1 \cdot 2}} \right]$$

$V_{DD} = ?$

$$\frac{\partial I_O}{\partial V_{DD}} = 0$$

~~X~~, T, P

From (3)

$$(\sqrt{I_O})^2 - \frac{1}{R} \sqrt{\frac{2}{\beta_1}} \sqrt{I_O} - \frac{V_T}{R} = 0$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1, b = -\frac{1}{R} \sqrt{\frac{2}{\beta_1}}, c = -\frac{V_T}{R}$$

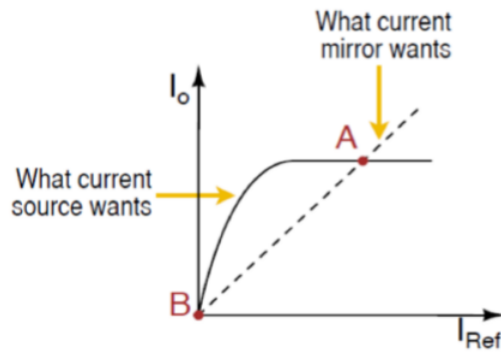
$$\Rightarrow \sqrt{I_O} = \frac{1}{R} \sqrt{\frac{2}{\beta_1}} + \sqrt{\frac{1}{R^2} \frac{2}{\beta_1} + \frac{4 V_T}{R}}$$

2(1)

In this expression V_{DD} has no direct or indirect role so we can say that the I_o is insensitive to the variations in V_{DD}

i.e. $\boxed{S_{V_{DD}}^{I_o} = 0}$

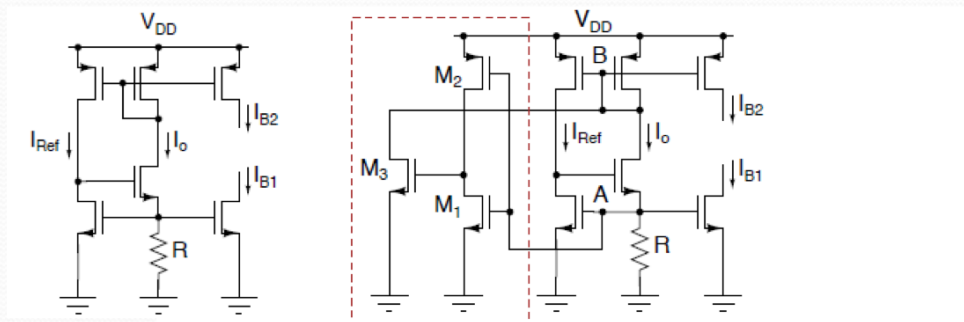
Note:- The only issue with this topology is we are ok with point A but not with point B



∴ In order to avoid B we have additional circuitry added to the circuit.

తేవోండ్ తల్లి పన్నెం
EBIT, Unibo

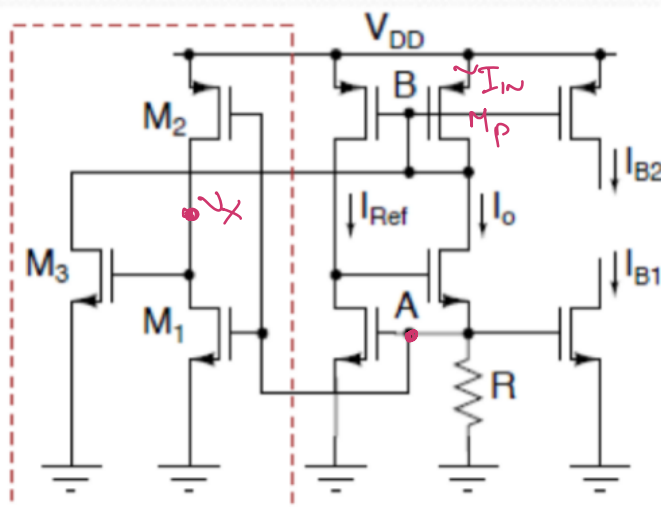
Self-Biasing: VT reference and Start-up Circuit



- In the undesired operating point, I_o would be zero pulling A to ground.
- The inverter formed by M_1 and M_2 will turn on, followed by M_3 , pulling B down so current would flow in the circuit.
- Need to make sure the start-up circuit does not interfere with normal operation.
 - Ensure that M_3 turns off properly by making M_1 stronger than M_2 .

EK

27



IF $I_o = 0$

- Then the voltage across R is also equal to zero
- Then $V_A = 0$ which is the i/p to the inverter with M_1 & M_2
- For $V_A = 0$, M_2 turns ON & $V_x = V_{DD}$ which

intern turns ON M_3
and pulls the ^{voltage} ^{node} value of B to zero

so that M_p turns on and I_O will reach saturation value and operate at the correct operating point.

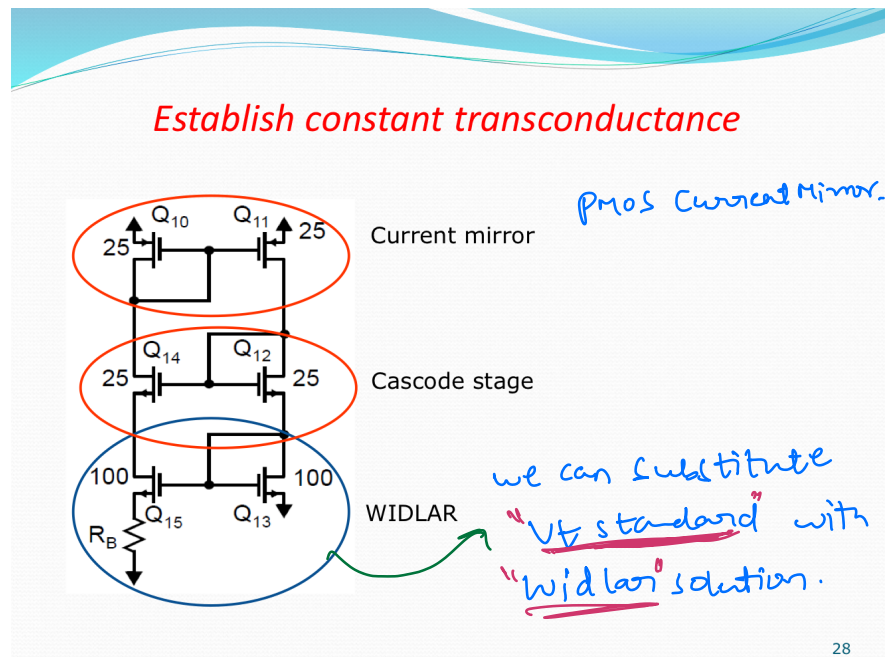
→ soon after I_O value reaches saturation

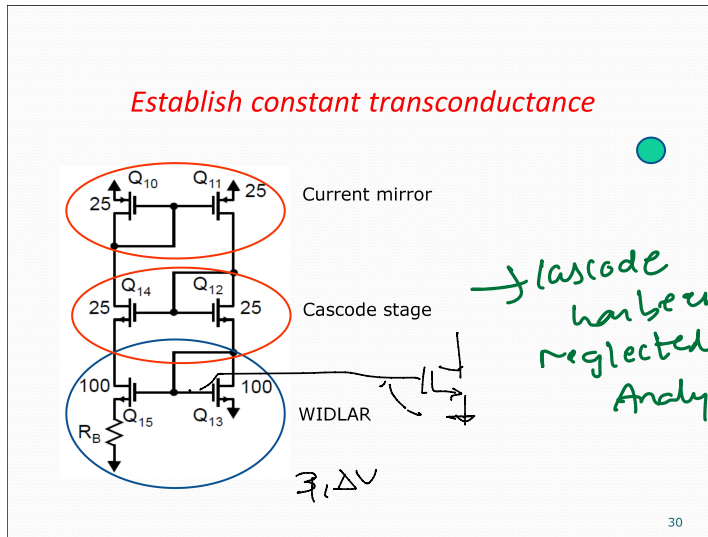
V_A is no longer zero

$$V_A = I_O R$$

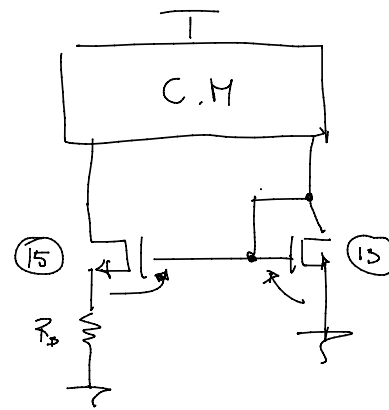
Then V_A which is applied to the inverter M_2 turns off and thereby M_3 also turns off.

* All The solutions we have studied till now can be applied to have a perfect current mirror.





cascode has been neglected in Analysis



$$V_{GS13} = V_{GS15} + I_{D15} \cdot R_B$$

$$V_T + V_{OV13} = V_{OV15} + V_T + I_{D15} \cdot R_B$$

$$V_{OV} = \sqrt{\frac{2I}{\beta}}$$

$$\sqrt{\frac{2I_{13}}{\beta_{15}}} = \sqrt{\frac{2I_{15}}{\beta_{15}}} + I_{15} \cdot R_B \quad ; \quad I_{15} = I_{13}$$

$$\frac{1}{I_{13}} \frac{\sqrt{2I_{13}}}{\sqrt{\beta_{15}}} = \frac{\sqrt{\frac{2I_{13}}{\beta_{15}}} + I_{13} \cdot R_B}{I_{13}}$$

$$\frac{\sqrt{2}}{\sqrt{\beta_{15}} \cdot \sqrt{I_{13}}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{\beta_{15}} \cdot \sqrt{I_{13}}} + R_B$$

$$\frac{2}{\sqrt{2\beta_{15} \cdot I_{13}}} = \frac{2}{\sqrt{\beta_{15} \cdot I_{13}}} + R_B$$

$$\beta_{15} = \left(\beta' \frac{W}{L} \right)_{15} \frac{\frac{W}{L}_{13}}{\frac{W}{L}_{15}} = \beta_{13} \cdot \frac{\frac{W}{L}_{15}}{\frac{W}{L}_{13}}$$

$$[g_{m15}] = \frac{2 \left[1 - \frac{\sqrt{\frac{W}{L}_{13}}}{\sqrt{\frac{W}{L}_{15}}} \right]}{R_B} \quad \leftarrow V_{DD}$$

$$g_{m13} = \sqrt{2\beta_{13} \cdot I_{13}}$$

$$\frac{2}{\sqrt{2\beta_{15} \cdot I_{13}}} \left[1 - \frac{\sqrt{\frac{W}{L}_{13}}}{\sqrt{\frac{W}{L}_{15}}} \right] = R_B$$

→ Establishing a constant transconductance for a desired transistor is too difficult

to have, the above configuration allows
us to achieve that

by

Revanth Reddy Pannala

EBIT, Unibo